Inequalities — problem 7.77

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Three lines are drawn through a point O inside a triangle with area S so that every side of the triangle is cut by two of them. The lines cut out of the triangle three triangular pieces with common vertex O and areas S_1, S_2, S_3. Prove that

$$\text{(a) } \frac{1}{S_1}+\frac{1}{S_2}+\frac{1}{S_3}\geq\frac{9}{S}, \qquad \text{(b) } \frac{1}{S_1}+\frac{1}{S_2}+\frac{1}{S_3}\geq\frac{18}{S}.$$